

The Power of Preprocessing in Image Classification: A Case Study with Breast Cancer Images

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Abstract

Leveraging machine learning techniques for image classification, particularly in the context of breast cancer detection, requires meticulous data preparation and a robust dataset to ensure accurate model performance. This document outlines the preprocessing steps and data augmentation techniques applied to a breast cancer image dataset, enhancing the model's ability to generalize and perform well in real-world scenarios, independent of variations in image quality and conditions.

1. Introduction

TODO

2. Data Augmentation

In this section, we describe the techniques applied to prepare the images in the database before their use in the machine learning model. The steps follow modern practices of data augmentation and image normalization, performed with the help of the **TensorFlow** and **Keras** libraries, in order to improve the model's generalization and increase robustness against common variations in real-world environments.

3. Data Augmentation

To simulate natural variations and enrich the training set, random transformations were applied to the images using Keras's **ImageDataGenerator** module. The chosen transformations aim to preserve the semantics of the image (i.e., keep the target objects recognizable) while introducing variations in:

- Horizontal and vertical translation: small changes in the positioning of the object;
- Zoom: controlled zooming in and out of the image;
- Rotation: changes in the viewing angle of the image around its center;
- Shear: diagonal distortions simulating perspective;
- Horizontal flip: simulates objects seen from opposite sides.

The corresponding data augmentation code was defined as:

```
from tensorflow.keras.preprocessing.image import ImageDataGenerator

train_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=10,
    width_shift_range=0.1,
    height_shift_range=0.1,
    shear_range=0.1,
    zoom_range=0.1,
    horizontal_flip=True,
    fill_mode='nearest'
)
```

4. Image Preprocessing

After applying the data augmentation transformations, the images are prepared for the model. This preparation includes:

- **Image resizing:** all images were resized to fixed dimensions (e.g., 224x224 pixels), ensuring consistency in the network input;
- **Pixel normalization:** pixel values were scaled to the $[0, 1]$ range using the `rescale=1./255` parameter, which facilitates the training of deep neural networks;
- **Batch conversion:** images were organized into batches for efficient GPU feeding during training.

Example of loading images with preprocessing applied:

```
train_generator = train_datagen.flow_from_directory(
    'dataset/train',
    target_size=(224, 224),
    batch_size=32,
    class_mode='categorical'
)
```

This strategy automates steps that in previous projects were performed manually, such as conversion to grayscale, adaptive thresholding, and noise removal with median filters.